

CSI WiFi Gesture Quality Recognition

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1 Introduction

Wireless sensing has been widely studied for its applications in gesture recognition in recent years. Research has focused on how human movement can be detected through its impact on wireless signals in order to classify the movement into specific gestures. This project aims to expand beyond classification into the field of assessment.

Certain applications benefit from information beyond classification, by instead learning about the gesture being performed. Specifically, in the field of fitness including the gym, sports, physical therapy, and other forms of exercise, the importance of maintaining form is evident. Focusing on the gym, maintaining form while you work out leads to increased effectiveness of the workout while also helping to prevent injury. Existing approaches to assist in maintaining form including vision based systems and wearable devices, can be expensive and inconvenient. To address this, we built a system to assess the quality of form while performing bicep curls through the use of wireless signals.

Our system derives quality assessment through the use of WiFi CSI signals. The system will rely on two ESP32 microcontrollers to collect CSI data, which will be processed into a time-frequency spectrogram representation, from which motion related features will be extracted and used to evaluate the quality of the curl being performed.

The goal of this work is to determine if the pipeline can output a quality assessment of the curl. The final output of the pipeline will include a quality score for the curl, along with potential feedback for improvement of the gesture. Successful results will demonstrate the potential for assessing gesture quality with CSI data, showing advancement into the field of gesture assessment is possible.

2 Literature Review

The works referenced in this section are from the Widar3.0 publications recommended by Professor Roshan. These works show that wireless sensing systems using WiFi Channel State Information (CSI) can be effective for capturing and classifying human movements. The works we analyzed from this conference include Widar 3.0, WiDance, and WiTrack.

Widar 3.0 demonstrates that a recorded CSI series can be processed into a body coordinate velocity profile, which is then analyzed to perform gesture classification. Through a series of signal processing techniques, Widar 3.0 takes raw CSI data and extracts motion related features as a BVP for use in classification. Widar 3.0's work demonstrates that it is possible to learn about motion through CSI data to differentiate between different gestures.

WiDance extracts Doppler features from CSI data, and uses them to infer the direction of motion for interactive exergames. WiDance's pipeline performs processing to turn raw CSI into the time-frequency domain in the form of a spectrogram. They highlight how Doppler shifts caused by human movement represented in the spectrogram can be used to analyze and classify motion patterns. Their demonstration of analyzing spectrograms directly influenced our own pipeline for creating and extracting features from spectrograms.

WiTrack introduced the foundational technique of isolating motion-related reflections by subtracting static ambient noise from raw CSI signals. Their methodology demonstrates that by filtering out environmental interference, it is possible to highlight the subtle signal variations specifically caused by human movement. This process of background subtraction serves as a critical precursor for accurate localization and tracking. WiTrack's approach to signal purification directly informed our own preprocessing pipeline, establishing the baseline required to extract clean, motion-centric data from noisy environments.

From these works it is shown that CSI data can be used to learn about and classify human motion. These works demonstrate pipelines for processing CSI data into motion related forms for the use of gesture classification. We will build upon the techniques of these works in order to go beyond classification and assess our hypothesis of gesture quality evaluation.

3 Technical Idea

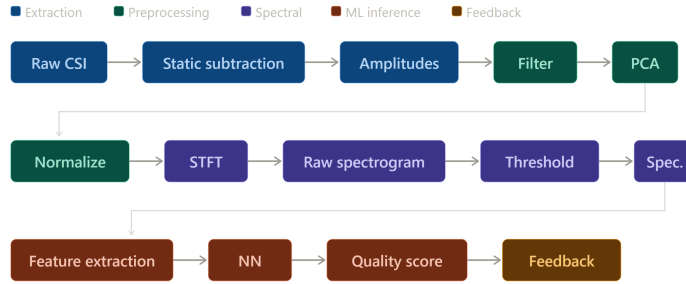
To evaluate the feasibility of gesture quality assessment, we designed a system that evaluates the quality of a bicep curl using WiFi CSI data. The key theory supporting our concept is that we can extract meaningful information about the gesture from the changes in CSI signals while the gesture is performed. The system has two parts, the hardware setup and ml pipeline processing the CSI data.

3.1 Hardware

The hardware for our system consists of two ESP32 microcontrollers. One ESP32 constantly broadcasts signals, while the other receives these signals and records CSI data for use in the pipeline. The two microcontrollers are spaced approximately 1 meter apart, while a user performs bicep curls between them. 5 seconds of CSI data will be recorded to provide time before and after the curl ensuring the full curl and signal variations are captured. For this work, only bicep curls are performed, allowing the focus to shift from gesture classification to quality evaluation.

3.2 Pipeline Overview

Our pipeline processes CSI data into an evaluation score with potential feedback, supported in theory by math. Outline:



Our pipeline is centered around the concept of the Doppler effect, with the belief that we can extract useful information from Doppler frequency shifts to evaluate the quality of the gesture. Doppler frequency shifts are caused by the movement of reflecting objects in the signal's path; in the case of gesture recognition, they are caused by the movement of body parts and can be represented as:

$$f_d = \frac{v}{\lambda} \quad (1)$$

Where v is the velocity of the moving object, and λ is the wavelength of the signal. This formula confirms that Doppler shifts are related to motion, specifically the velocity of motion. Therefore, by analyzing these shifts we should be able to learn features about the curl being performed and use these features to evaluate the quality of the curl. However, Doppler shifts aren't readily apparent in raw CSI data. In reality, raw CSI measurements are difficult to interpret, and are affected by multipath propagation where signals reflect off of multiple surfaces in the environment. CSI can be modeled as:

$$H(f, t) = H_s(f) + \sum_{k=1}^K a_k(t) e^{-j\phi_k(t)} \quad (2)$$

Where the total CSI $H(f, t)$ is equal to the sum of static paths $H_s(f)$, and the sum of the K dynamic paths for amplitude $a_k(t)$ and phase shift $\phi(t)$.

3.3 Amplitude Extraction

Before we can utilize the Doppler effect, multipath influence must be accounted for. Inspired by WiTrack, we recorded a baseline gesture with minimal dynamic interference. With dynamic paths controlled to a minimum, it is expected the recorded CSI values represent solely the static environment. These values are stored and used in our pipeline so that when we record CSI data, the first step is to subtract the static CSI values, essentially removing static multipath leaving only the dynamic values.

During experiments, we also maintain a controlled environment with minimal external movement, for the purpose of preventing any dynamic paths other than ones resulting from the performed gesture. While we can't completely prevent unwanted dynamic paths, it is assumed their impact is minimal enough to be ignored, and the gestures paths will be responsible for recorded shifts.

The complex values of CSI data are unusable in their raw form, so we extract the amplitudes using the formula:

$$|H(f)| = \sqrt{\text{Real}^2 + \text{Imaginary}^2} \quad (3)$$

Where the magnitude of raw CSI values is used to find the amplitude. While we could also extract phases using a similar formula, they will be ignored as our previous works such as Widar 3.0 have claimed phases are noisy and hard to work with.

3.4 Preprocessing

The raw amplitudes produced from CSI values like the ones in Figure 1A still contain noise, large quantities of data, and are not yet able to show Doppler shifts. A series of preprocessing steps including filtering, pca, and normalization are used to reduce the raw amplitudes into a single amplitude vs time series.

Filtering is done first, applying a Butterworth Bandpass Filter to each subcarrier over the time domain. Filtering is done to reduce low and high frequency noise. The filtered amplitudes in Figure 1B show a more tightened range of amplitudes.

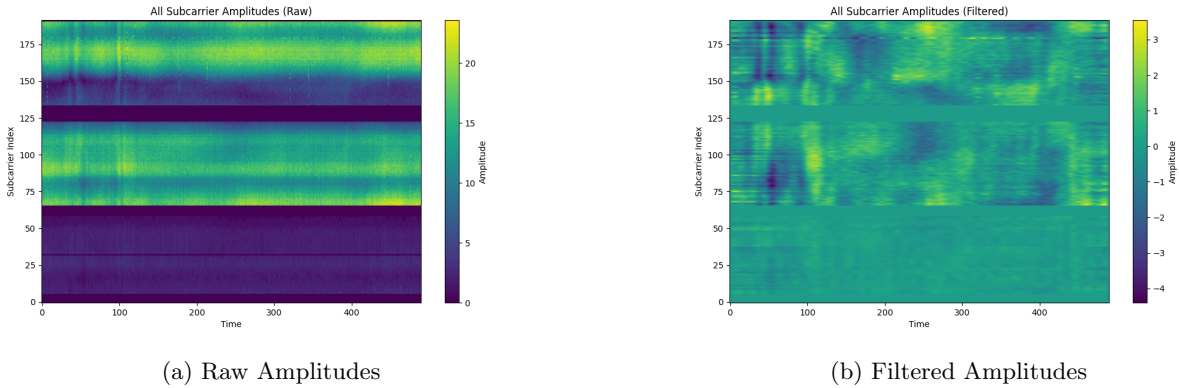
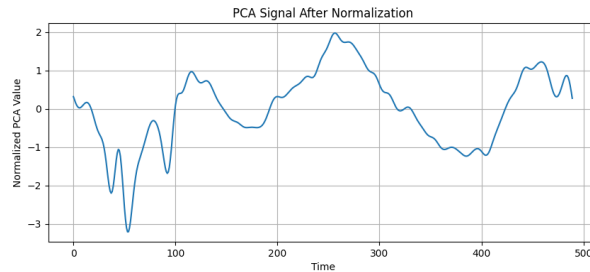


Figure 1: Amplitudes before and after filtering.

Next, we utilize Principal Component Analysis (PCA). Rather than simply averaging subcarriers which can destroy the signal, PCA identifies the direction of maximum variance across all subcarriers. We extract the first principal component (PC_1), which captures the dominant energy of the bicep curl, giving a single PC series which we normalize:



3.5 Spectrogram Generation

Once down to a single series, an STFT is applied to convert the data into a frequency vs time spectrogram. In spectrogram form, we can begin analyzing features created by Doppler shifts. An example of a raw spectrogram is shown in Figure 2A below on the left. The raw spectrogram still has some levels of noise, so we apply thresholding to remove noise below a certain percentile of the energy distribution. This removes all remaining noise while leaving behind the clean signal spectrogram as shown in Figure 2B on the right.

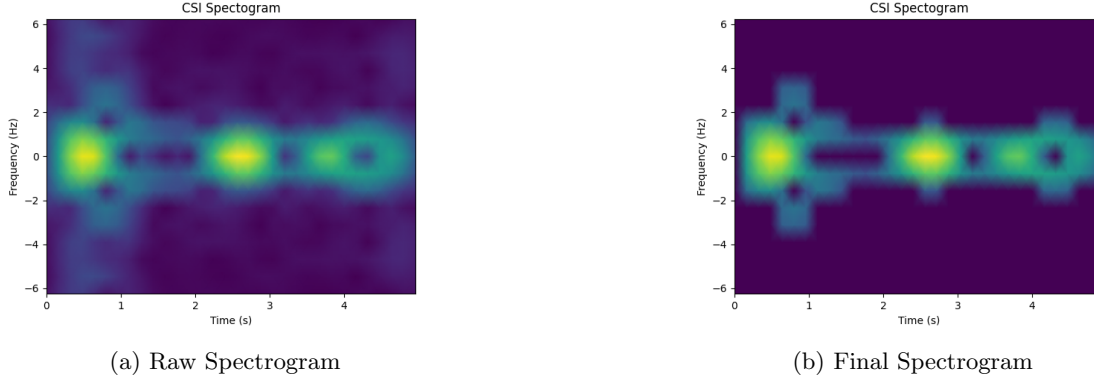


Figure 2: Spectrogram before and after thresholding.

3.6 Feature Extraction & Neural Network

To extract motion related features from spectrograms, a few techniques are employed. We first extract energy related features including the total energy in the spectrogram, the standard deviation of energy distribution, and the energy in low, middle, and high frequency bands.

Besides energy, we also create an energy envelope, which entails a line tracing the spectrogram along the points where a specific amount of energy is below the line like in Figure 3A. A sample resulting line is shown in Figure 3B.



Figure 3: Process and results for energy envelopes.

From the envelope line we extract a few features relating to duration, peak frequency, mean frequency, duration above certain frequencies, and the mean above these frequencies. Extracted features are fed to a Neural Network which will output a quality score between 0 and 1. To train the network, data containing various forms of curls given manually assigned quality scores as labels for supervised learning are used. These scores serve as the ground truth, establishing what is considered a good repetition. The model will learn to turn extracted features into a quality score. This score, along with potential feedback derived from the features will be the final output. The quality score will demonstrate the feasibility of evaluating gesture quality and providing feedback using CSI data.

4 Implementation Details

4.1 Hardware and Data Acquisition

Our system is built upon an edge-computing architecture utilizing two **ESP32-S3** microcontrollers acting as a dedicated transmitter (TX) and receiver (RX) pair. The nodes are positioned 1 meter apart in a line-of-sight (LoS) configuration to minimize multi-path fading from external objects. Gestures are performed in line with the signals path.

We leverage the **ESP-IDF** framework to extract raw Channel State Information (CSI) from 192 subcarriers. Data collection is orchestrated via a custom Python script that manages the serial interface, triggering synchronized 5-second CSI bursts for each exercise repetition. These raw packets are converted into structured CSV format using the **CSI-Toolkit** parser and centralized on a single laptop for processing.

4.2 Signal Processing

To transform noisy RF signals into actionable motion features, we implement a multi-stage digital signal processing (DSP) pipeline:

- **Static Noise Removal:** Adopting techniques introduced by **WiTrack**, we subtract a pre-recorded static ambient noise profile from the raw complex CSI. This isolates motion-related reflections by removing the static multipath components of the environment.
- **Filtering:** The signal is processed through a 4th-order **Butterworth bandpass filter** (0.2 Hz – 7 Hz) to eliminate high-frequency thermal noise and low-frequency DC offsets.
- **Dimensionality Reduction:** We apply **Principal Component Analysis (PCA)** to the amplitude of the complex signal, retaining the first principal component to capture the maximum variance caused by body movement. The resulting component is passed through Z-score Normalization to scale the signal, ensuring consistent feature scaling regardless of the user’s distance from the sensors.
- **Spectrogram Analysis:** We perform a **Short-Time Fourier Transform (STFT)** to generate a time-frequency representation. To enhance the signal, we apply a 97th-percentile energy threshold to mask background noise.

4.3 95% Cumulative Energy Envelope

A core technical contribution of our implementation is the **95% Cumulative Energy Envelope**. Unlike standard peak detection (argmax), which is susceptible to noise jitter, our pipeline calculates the frequency boundary where 95% of the spectral energy is contained. This provides a stable, continuous representation of the movement’s true frequency peak over time, which serves as the primary input for feature extraction, alongside energy related features. Energy features include the total, std, and 3 bands. The 3 bands low, middle, and high encompass feature ranges $f < 1$, $1 \leq f < 4$, and $f \geq$ respectively.

4.4 Model Architecture and Training

The classification and scoring engine is a **PyTorch-based feed-forward neural network**. The architecture consists of an input layer (10 features), a 16-node hidden layer with **ReLU** activation, and a single-node output layer with a **Sigmoid** activation function. This configuration allows the model to output a continuous score between 0 and 1, representing exercise quality.

4.5 Testing and Validation

We tested the system using a labeled dataset of 30 exercise repetitions spanning six categories: Full/Half range of motion and Fast/Normal/Slow speeds. The model was trained using the **Adam optimizer** (learning rate = 0.01) for 300 epochs, minimizing Mean Squared Error (MSE).

Testing was performed on a 20% stratified hold-out set. The system achieved **83% accuracy** within a 15% error margin. Specifically, the average prediction error for "Full Reps" was 0.0433, and 0.1600 for "Half Reps." This validation proves the system’s ability to accurately differentiate between correct and incorrect form and provide real-time qualitative feedback (e.g., "Perfect!" vs. "Remember to do a full range of motion").

5 Results

The system’s performance was evaluated using a 20% stratified hold-out test set, comprising unseen repetitions across the six defined categories (Full/Half range of motion executed at Fast/Normal/Slow speeds). The primary metric for evaluation was the average prediction error, representing the absolute difference between the neural network’s predicted quality score and the manually labeled ground truth score (ranging from 0.0 to 1.0).

5.1 Quantitative Performance

Overall, the pipeline demonstrated strong predictive capabilities, achieving an **83% accuracy rate**. In this context, a prediction was considered accurate if it fell within a strict 15% error margin (± 0.15) of the actual ground truth label. This continuous scoring capability confirms that the system can reliably quantify the degree of correctness rather than simply classifying a gesture as "good" or "bad".

Exercise Variation	Average Prediction Error
Full Range of Motion	0.0433
Half Range of Motion	0.1600

Table 1: Model performance metrics across different ranges of motion.

5.2 Key Observations

- **Impact of Range of Motion:** As highlighted in Table 1, the system exhibited higher precision when evaluating full repetitions. Full bicep curls generate more sustained and pronounced Doppler frequency shifts over a longer duration. This results in a wider, more distinct 95% Cumulative Energy Envelope, providing the neural network with highly defined features. Conversely, half repetitions produce shorter, less energetic signal variations, making subtle form discrepancies slightly harder to distinguish from background noise.
- **Consistency Across Speeds:** The model maintained consistent performance across varying execution speeds. However, it was observed that "Slow" repetitions occasionally exhibited minor fluctuations in the predicted score. Because the Doppler shift is directly proportional to velocity ($f_d = \frac{v}{\lambda}$), slower movements resulted in lower-frequency shifts that hovered closer to the lower boundary of our Butterworth filter, marginally reducing feature clarity compared to "Normal" and "Fast" speeds.
- **Actionable Feedback System (Qualitative Translation):** The core utility of the model's continuous numerical output lies in its translation to actionable, real-time coaching cues. Based on the quality score predicted by the neural network, the system maps the gesture into four distinct feedback tiers:
 - **Score > 0.90 (Perfect!):** Indicates a full-range, controlled-speed repetition. The extracted Doppler spectrogram matches the ideal pattern closely, triggering the positive cue: *"Perfect! Keep it up."*
 - **Score 0.70 – 0.89 (Speed Adjust):** Indicates generally good form, but the tempo is off (e.g., executing the curl too fast or too slow). This triggers the tempo-adjustment cue: *"Almost there! Slow down / Speed up a little."*
 - **Score 0.40 – 0.69 (Range Issue):** Indicates a half-rep where the user's arm is not completing the full curl arc. This triggers the corrective cue: *"Remember to do a full range of motion,"* advising the user to focus on extending fully at the bottom and contracting at the top.
 - **Score < 0.40 (Seek Help):** Indicates a fundamental form breakdown where the movement pattern is unrecognizable or severely incomplete. This triggers the critical safety cue: *"Seek help with form!"*

6 Discussion

The results from our implementation successfully validate our initial hypothesis: it is feasible to utilize WiFi CSI data beyond simple classification to directly evaluate the quality of a specific human movement. By achieving an 83% accuracy within a 15% error margin and demonstrating low average prediction errors for differing ranges of motion (0.0433 for full repetitions and 0.1600 for half repetitions), our system proves capable of effectively distinguishing between proper and improper bicep curl form. The continuous quality score generated by the neural network, driven by extracted features like the 95% Cumulative Energy Envelope, confirms that subtle variations in motion velocity and energy distribution can be reliably mapped to a qualitative performance metric.

Despite these promising results, there are notable pitfalls and limitations in our current work. First, the model was trained and validated on a highly restricted dataset of only 30 repetitions. Such a small sample size restricts the neural network's ability to generalize across different users with varying body types, execution speeds, and minute variations in form. Second, our signal processing pipeline relies heavily on a highly controlled environment. The static noise removal technique assumes that the exercising user is the only moving entity; in a practical setting like a public gym, dynamic multi-path interference from other individuals would introduce significant noise. Furthermore, our experiments were limited to a strict line-of-sight (LoS) configuration for a single exercise, and phase data was intentionally discarded due to its high noise floor, potentially leaving valuable spatial and directional information untapped.

Future work should address these limitations to build a more robust and scalable wireless sensing system. A primary direction is expanding the dataset to include hundreds of repetitions from diverse users and evaluating the pipeline on a wider variety of exercises, such as squats or shoulder presses. Additionally, advanced signal processing techniques, such as conjugate multiplication across antennas, could be explored to successfully denoise and integrate phase data into the feature

set for better spatial resolution. Finally, transitioning the system to handle dynamic, multi-person environments—perhaps through advanced source separation algorithms—will be critical for deploying this technology as a practical, real-time fitness assessment tool in everyday settings

7 Work Distribution

Harshvardhan: Beginning of Pipeline, Final Presentation, Demo Day Poster

Joshua: Hardware Setup, All Pipeline Implementation

Both members discussed the overall project design at each stage and collaborated on the data collection, pipeline implementation and final report.

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